

Formal Comparison of Critical Behavior: Plausible (GSS) vs. Random (ER) Topologies

A Theoretical Evaluation of Topological Destruction

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Abstract

This report provides a formal mathematical and physical evaluation of the phase transition behaviors observed in a social contagion model simulated over empirically grounded plausible networks (GSS) versus appropriately randomized variants (Erdős-Rényi, ER). By analyzing the critical susceptibility exponents (γ) and the macroscopic order parameter macroscopic jumps ($\Delta\Phi$), we demonstrate that the destruction of homophilic structural clustering results not in a partial degrading of network diffusion capacity, but in a complete thermodynamic collapse of the universality class. We establish that plausible networks exhibit continuous (second-order) phase transitions belonging to the Mean-Field Ising universality class ($\gamma \approx 1$), while random networks collapse into trivial, discontinuous (first-order) cascades.

1 Introduction and Thermodynamic Framework

The study of collective social contagion can be mapped onto the formalisms of statistical physics by defining an order parameter $\Phi \in [0, 1]$, representing the global proportion of adopters, and an external control parameter MSP (Maximum Social Proximity), acting as the systemic thermal driving force (an inverse temperature analogue) against the intrinsic rigidity defined by individual thresholds (IUL).

In the thermodynamic limit ($N \rightarrow \infty$), the system's susceptibility χ , which measures the responsiveness of the order parameter to infinitesimal shifts in the control parameter, is governed by the Fluctuation-Dissipation Theorem (FDT):

$$\chi \propto N \cdot \text{Var}(\Phi) \tag{1}$$

Near a critical boundary MSP_c , if the system undergoes a continuous (second-order) phase transition, the susceptibility diverges according to a universal power law:

$$\chi \sim |MSP - MSP_c|^{-\gamma} \tag{2}$$

The critical exponent γ exclusively defines the “Universality Class” of the system. Its value depends strictly on the fundamental symmetries and the spatial dimensionality of the interactions, ignoring the micro-level nuances of the specific model.

2 Plausible Networks (GSS): The Richness of Continuous Criticality

For the plausible geometries imputed holding the empirical General Social Survey (GSS) parameters, diffusion exhibits a macroscopic continuous phase shift. Over the non-biased analogue slice (IUL ≈ 0.200), the system yields an exponent of:

$$\gamma_{GSS} \approx 0.977 \pm 0.01 \quad (3)$$

This value aligns extraordinarily well with the theoretical expected value for the **Mean-Field Ising Universality Class** ($\gamma_{MF} \equiv 1.0$).

2.1 The Structural Meaning of $\gamma \approx 1$

The mathematical richness of possessing a $\gamma \approx 1.0$ is immense. It proves that the human homophilic topology is capable of sustaining *scale-free* fractal fluctuations. In the critical zone, the network contains highly connected homophilic “echo chambers” (clusters of localized resistance or facilitation) that act as macroscopic domain walls.

These domain walls impose a structural friction that forces the order parameter Φ to grow continuously. Avalanches of adoption occur at all possible size scales (power-law distribution of cascades), allowing the variance to act smoothly as a predictive precursor to systemic phase change. The universality class validates that societal coordination is not an algorithmic artifact, but a fundamental emergent property of complex systems behaving as a unified whole.

3 Topological Destruction (GSS-ER) and the Loss of Universality

To evaluate the pure causal effect of the empirical structure, the GSS topologies were deliberately destroyed via random rewiring, preserving only the macroscopic density whilst imposing a Poissonian degree distribution (Erdős-Rényi, ER).

The destruction of the structure is not partial; it is **total** from a thermodynamic perspective. For the GSS-ER control networks, the susceptibility exponent inflates erratically:

$$\gamma_{ER} \approx 3.7 \text{ to } 4.6 \quad (4)$$

These values do not belong to any formal Universality Class in statistical physics. More critically, for the super-critical regime ($MSP > MSP_c$), the variance of the order parameter collapses to mathematically exactly zero ($\text{Var}(\Phi_{ER}) = 0$).

3.1 Collapse into First-Order Discontinuity

The ER results prove the absence of a continuous phase shift. By destroying the local clustering, the ER network drastically reduces the average path length and eliminates the echo-chamber domain walls.

Consequently, once the energetic boundary MSP_c is marginally breached, the random network cannot offer any local friction capable of stopping the contagion. Every identically initialized Monte Carlo seed triggers an immediate, unmitigated global cascade ($\Phi \rightarrow 1$). This describes a **First-Order (Discontinuous) Phase Transition**. Because the jump is instantaneous, there are no scale-free fractal fluctuations, no gradual continuous divergences, and the susceptibility drops to zero post-jump. The calculated γ_{ER} exponents are simply computational artifacts of fitting an invalid continuous regression to a fundamentally discontinuous wall.

4 Conclusion: What is Lost?

The comparative analysis reveals that substituting an empirically plausible network with a random topology destroys much more than localized demographic realism; it extinguishes the entire thermodynamic character of the social system.

The richness lost is the **complexity** itself. Random networks render the population as a mathematically trivial conduit where information propagates identically and predictably. In stark contrast, the $\gamma \approx 1.0$ derived from the GSS matrices demonstrates that empirical social topologies uniquely construct a continuous structural tension between localized identity preservation (homophily) and systemic coordination. The continuous criticality relies *entirely* on the non-random arrangement of societal ties.

Therefore, we conclude that the capacity of a society to undergo rich, non-linear, self-organized behavioral transitions is exclusively scaffolded by its topological geometry. The network, quite literally, decides the fundamental laws of physics the societal collective will obey.